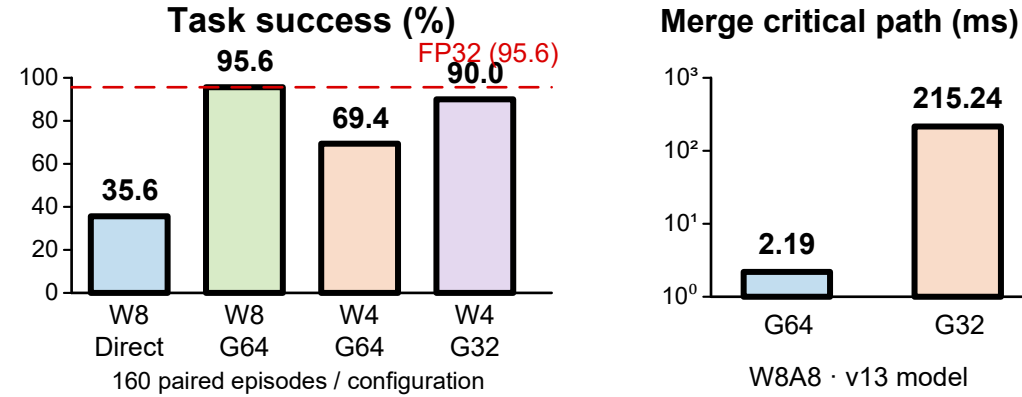


Challenges and hardware-aware solutions for low-bit VLA inference

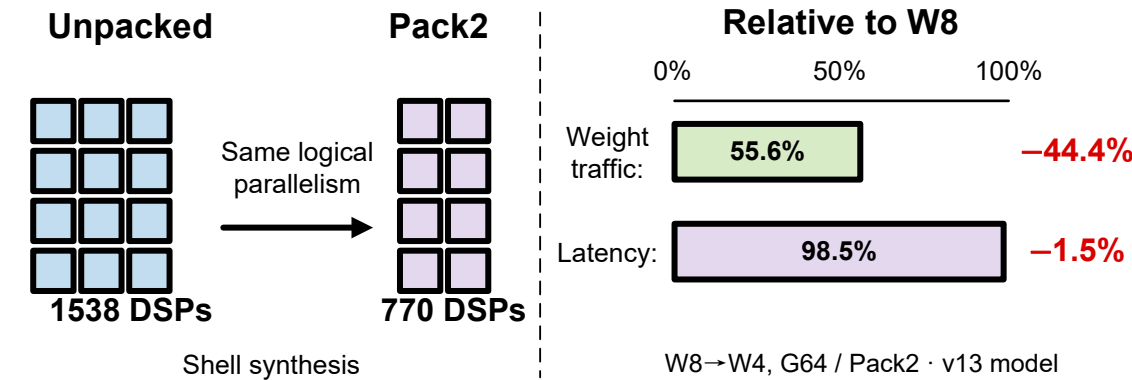
Challenges

C1: Quantization quality vs. grouping cost



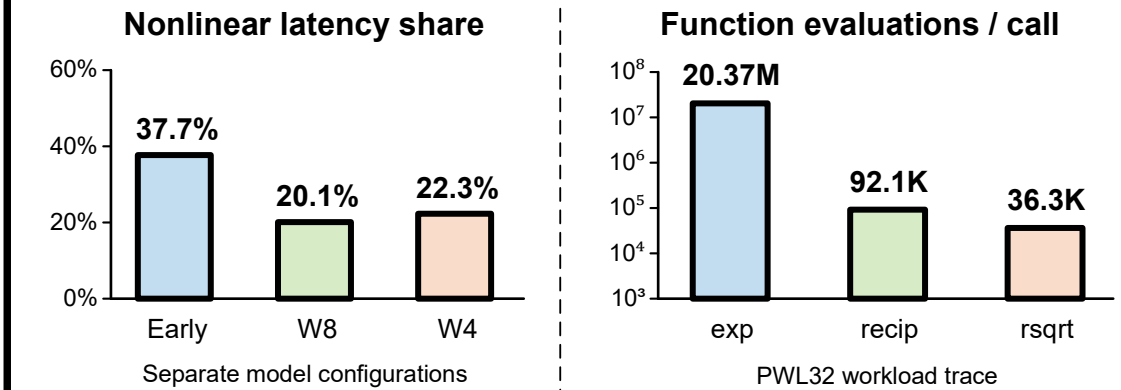
Finer groups improve quality but raise merge cost.

C2: Low-bit gains need hardware mapping



Weight compression alone gives limited system speedup.

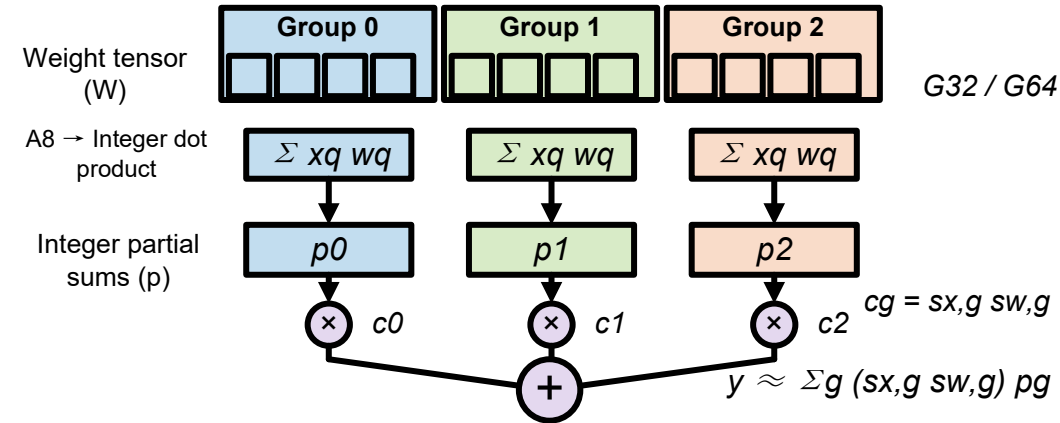
C3: Nonlinear cost and uneven demand



Nonlinear work remains costly and heterogeneous.

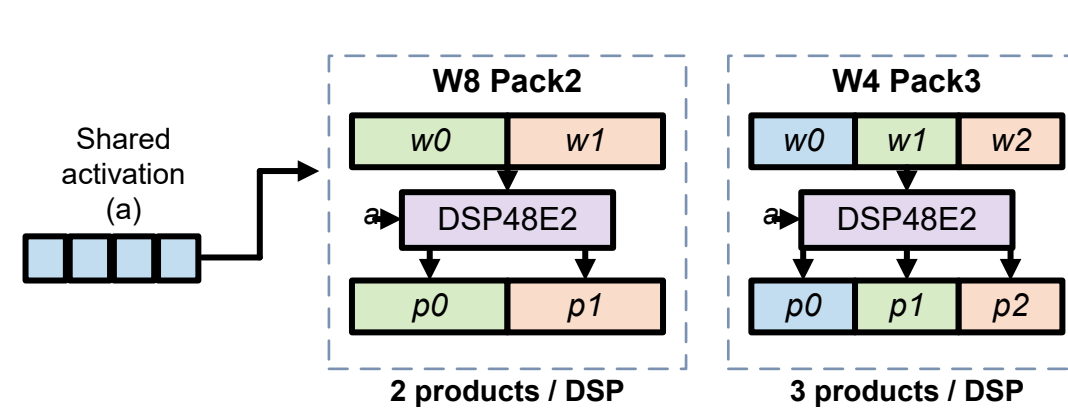
Methods

S1: Group-wise quantization and scaled merging



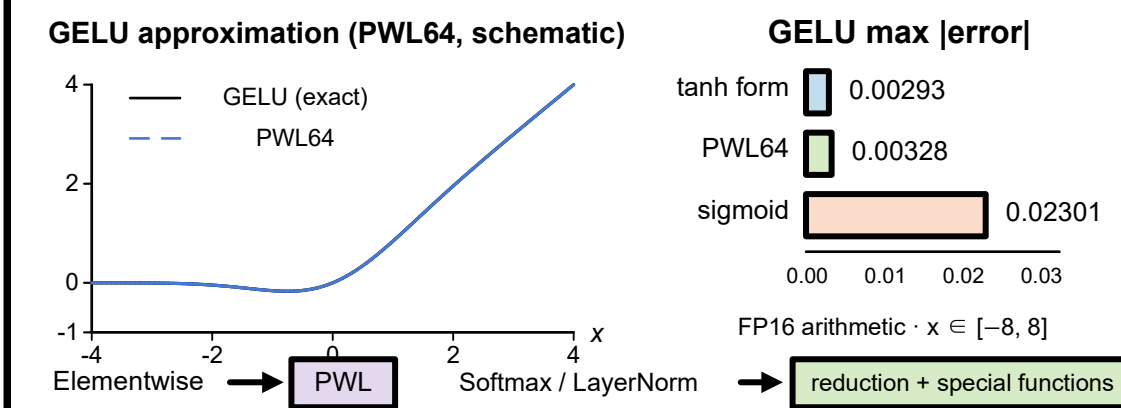
Preserve local numerical range with explicit scale handling.

S2: Shared-activation DSP packing



Increase arithmetic density within a fixed DSP budget.

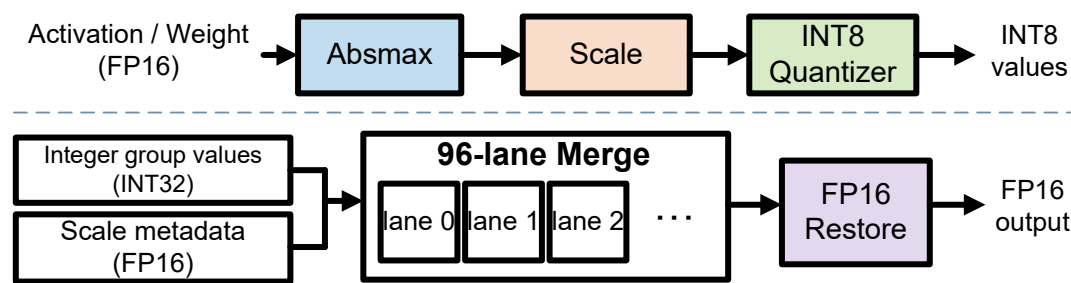
S3: Function-specific nonlinear evaluation



Use approximation and dedicated functions where appropriate.

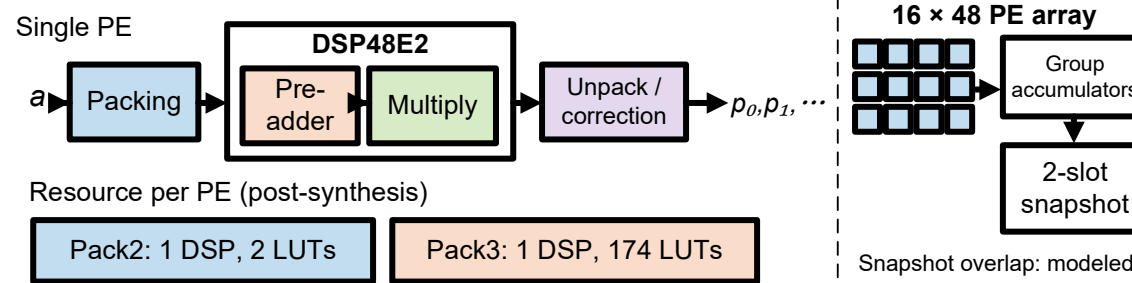
Hardware

H1: On-chip quantization and group merge



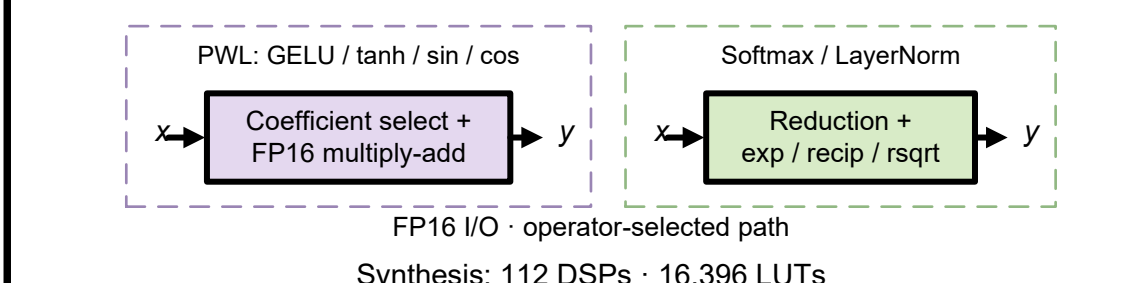
Pipeline grouped numerical processing.

H2: DSP-packed matrix engine



Trade correction logic for higher DSP compute density.

H3: Hybrid nonlinear unit



Combine PWL arithmetic with specialized nonlinear resources.